

Lab 3 – Answer Sheet

Arquitectura de Xarxes – 2025-2026 3rd quarter

Universitat Pompeu Fabra

Group members:

<u>Name</u>	<u>NIA</u>
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Submission: rename this file to `AX_lab3_NIA1_NIA2_NIA3.pdf` and submit before the next lab session.

Questions

Q1: What is the size of the Nginx image? What factors influence the size of a Docker image? Is a larger image preferable or a smaller one? Justify your answer.

Q2: What does each line of your Dockerfile do? If the build failed, describe what you found wrong and how you fixed it.

Q3: Why is `my-nginx` not significantly larger than `nginx`, even though you added a file to it? What are Docker image layers?

Q4: What is the purpose of each of the three default Docker networks (`bridge`, `host`, `none`)?

Q5: When you run a container without specifying `--net`, which network does Docker use by default?

Q6: What limitation does the default `bridge` network have compared to a user-defined network?

Q7: Both `curl http://172.18.0.2:80` and `curl http://localhost:8080` reach the same container. Explain why. What role does `--publish 8080:80` play?

Q8: What do the flags `-i` and `-t` mean in `docker exec -it`?

Q9: Why is the network interface output different inside the container compared to the host?

Q10: Can the host and the container be considered separate network entities? Justify your answer.

Q11: What happens to a directory created inside a running container when the container is stopped and a new one is started from the same image? Why?

Q12: Add a screenshot of the live log output below. What information does each log line contain?

Q13: What happened to the logs from the previous container after restarting? Is this a problem? Describe a real-world scenario where losing logs would have serious consequences.

Q14: What does `access.log` contain?

Q15: After stopping and restarting the container with a volume mount, are the previous log entries still there? Why?

Q16: What problem does the volume mount solve? What would happen without it?

Q17: Does communication between containers by IP work? Why?

Q18: Does communication by container name work? Why? Would this work on the default bridge network?

Q19: What happens when you try to reach `webserver2` from `webserver` after stopping it?

Q20: In a production environment, what mechanism would you use to avoid hardcoding container IPs?

Q21: You changed the content served by `webserver2` without rebuilding the image. What does this tell you about the relationship between images and containers? What is the conceptual difference between a Docker *image* and a Docker *container*?

Q22: The image pushed to Docker Hub is public by default. What are the security implications? In what cases would you use a private registry instead?

Q23 (Optional): Can `web` reach `web2` by service name in Docker Compose? Why?

Q24 (Optional): After running `docker compose down`, what happens to the containers, the network, and the volume mount?

Q25 (Optional): What is the main advantage of `docker-compose.yml` compared to individual `docker run` commands?